20261022 Quiz

for (MATH301) Modern Algebra I

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Exercise 4.5.13. Prove that a group of order 56 has a normal Sylow p-subgroup for some prime p dividing its order.

Solution. Since $56 = 2^3 \cdot 7$, Sylow's theorem gives $n_7 \mid 8$ and $n_7 \equiv 1 \pmod 7$, so $n_7 = 1$ or 8. If $n_7 = 1$, the Sylow 7-subgroup is normal.

If $n_7 = 8$, the eight subgroups of order 7 intersect pairwise only in the identity, because their order is prime. They therefore contain $8(7-1) = 48$ distinct nonidentity elements, all of order 7. The other eight elements consist of the identity and seven elements not of order 7. Any Sylow 2-subgroup has order 8 and cannot contain an element of order 7, so it is exactly this eight-element set. Consequently all Sylow 2-subgroups coincide: the unique Sylow 2-subgroup is normal.

□

Exercise 4.5.16. Let $|G| = pqr$, where $p < q < r$ are primes. Prove that G has a normal Sylow subgroup for at least one of p, q, r.

Solution. Sylow's theorem gives $n_r \mid pq$ and $n_r \equiv 1 \pmod r$. Because neither p nor q is congruent to 1 modulo r, $n_r = 1$ or pq. If $n_r = 1$, the Sylow r-subgroup is normal.

Suppose $n_r = pq$. Its pq distinct subgroups of order r have pairwise trivial intersections and contain $pq(r-1)$ nonidentity elements, leaving only $pq - 1$ nonidentity elements outside them. Also $n_q \mid pr$ and $n_q \equiv 1 \pmod q$. The possible divisors of pr are $1, p, r, pr$; since $p < q$, the value p is not 1 modulo q. Thus if $n_q > 1$, necessarily $n_q \geq r$. The distinct Sylow q-subgroups would then contribute at least $r(q-1)$ nonidentity elements, disjoint from the elements of order r. But

$$r(q-1) \geq rp > pq - 1,$$

since $q - 1 \geq p$ and $r > q$. This is impossible. Therefore $n_q = 1$, so a Sylow q-subgroup is normal.

□

Exercise 4.5.17. Prove that every group G of order 105 has a normal Sylow 5-subgroup and a normal Sylow 7-subgroup.

Solution. Since $105 = 3 \cdot 5 \cdot 7$, Sylow's theorem gives $n_5 \in \{1, 21\}$ and $n_7 \in \{1, 15\}$. If both were greater than 1, the distinct subgroups of orders 5 and 7 would contain $21 \cdot 4 + 15 \cdot 6 = 174$ distinct nonidentity elements. This exceeds $104 = |G| - 1$, so at least one of the two Sylow subgroups is normal.

First suppose $P_7 \trianglelefteq G$. Then G/P_7 has order 15. The number of its Sylow 5-subgroups divides 3 and is 1 modulo 5, hence is 1. Its unique Sylow 5-subgroup has a normal preimage

$K \trianglelefteq G$ of order 35. In K, the number of Sylow 5-subgroups divides 7 and is 1 modulo 5, hence is 1. This unique subgroup of order 5 is invariant under every conjugation by G, because G normalizes K and conjugation preserves subgroup orders; thus it is normal in G.

Conversely, if $P_5 \trianglelefteq G$, then G/P_5 has order 21. Its number of Sylow 7-subgroups divides 3 and is 1 modulo 7, so equals 1. Its preimage is again a normal subgroup K of order 35. The Sylow 7-subgroup of K is unique (its number divides 5 and is 1 modulo 7). It is consequently invariant under conjugation by all elements of G and normal in G. Hence both Sylow subgroups of G are normal.

□

Supplementary Exercise. Let G be a group with $|G| = 30$. Determine whether G can be simple. Justify your answer.

Solution. Suppose G were simple. Since $|G| = 2 \cdot 3 \cdot 5$, Sylow's theorem yields

$$n_5 \mid 6, \quad n_5 \equiv 1 \pmod 5 \qquad \Longrightarrow \qquad n_5 \in \{1, 6\},$$

and similarly $n_3 \in \{1, 10\}$. Simplicity rules out a unique normal Sylow subgroup, so $n_5 = 6$ and $n_3 = 10$. Distinct subgroups of prime orders have trivial pairwise intersections. Hence the Sylow 5-subgroups contribute $6 \cdot 4 = 24$ distinct elements of order 5, and Sylow 3-subgroups contribute $10 \cdot 2 = 20$ distinct elements of order 3. Counting the identity gives at least $1 + 24 + 20 = 45$ elements, contradicting $|G| = 30$. Therefore no group of order 30 is simple.

□

Exercise 5.1.18. Give examples of groups with the following properties.

(a) An infinite group in which every element has order 1 or 2.

(b) An infinite group in which all elements have finite order, with an element of order n for every positive n.

(c) A group with an infinite-order element and an element of order 2.

(d) A group containing an isomorphic copy of every finite group.

(e) A nontrivial group G with $G \cong G \times G$.

Solution.

(a) Let $G = \bigoplus_{n \geq 1} C_2$. This is infinite, and every element has finite support. Squaring coordinatewise gives the identity, so every nonidentity element has order 2.

(b) Take $G = \bigoplus_{n \geq 1} C_n$. Any element has only finitely many nonidentity coordinates, each of finite order, so its order is the least common multiple of finitely many finite numbers. The generator of the nth factor gives an element of order n.

(c) Take $G = \mathbb{Z} \times C_2$. The element $(1, 1_{C_2})$ has infinite order, while $(0, t)$ has order 2 for the nonidentity $t \in C_2$.

(d) Take $G = \bigcup_{n \geq 1} S_n$, with $S_n \leq S_{n+1}$ via permutations fixing $n + 1$. This union is a group. Cayley's theorem embeds each finite group H in $S_{|H|}$, hence in G.

(e) Take again $G = \bigoplus_{n \geq 1} C_2$. Splitting the coordinates into odd and even indices gives a group isomorphism $G \cong G \times G$; G is certainly nontrivial.

□

Exercise 5.2.10. Let A be a free abelian group of rank n and let $k > 0$. Show that A/kA is isomorphic to the direct product of n copies of $\mathbb{Z}/k\mathbb{Z}$, where $kA = \{ka : a \in A\}$.

Hint. Choose a basis of A and reduce each integer coefficient modulo k.

Solution. Choose a basis $a_1, \ldots, a_n$ of A. Every $a \in A$ has a unique representation $a = m_1 a_1 + \cdots + m_n a_n$ with integers m_i. Define

$$\begin{aligned} \varphi : A &\longrightarrow (\mathbb{Z}/k\mathbb{Z})^n, \\ \sum_{i=1}^{n} m_i a_i &\longmapsto ([m_1]_k, \ldots, [m_n]_k). \end{aligned}$$

Uniqueness of the coefficients makes the map well defined, and addition of coefficients shows it is a homomorphism. It is surjective because every tuple of residue classes has integer representatives. Moreover,

$$\varphi(a) = 0 \iff k \mid m_i \text{ for every } i \iff a \in kA,$$

so $\ker \varphi = kA$. By the First Isomorphism Theorem,

$$A/kA \cong (\mathbb{Z}/k\mathbb{Z})^n.$$

□